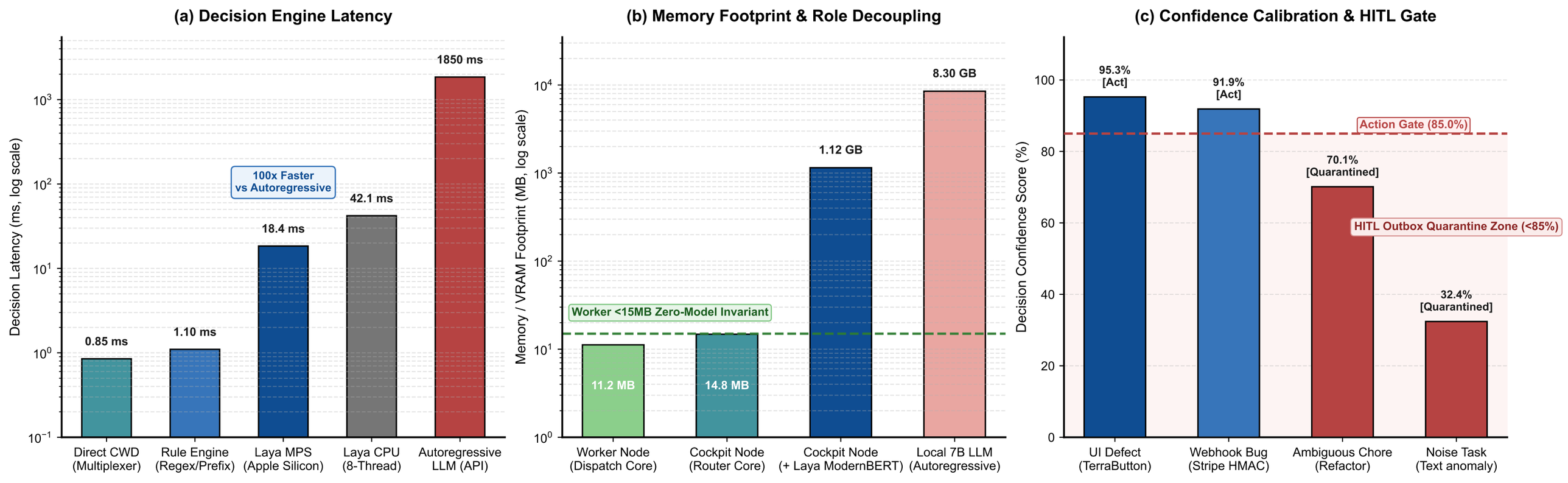
(a) Decision Engine Latency
Decision Latency (ms, log scale)
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
0.85 ms
1.10 ms
18.4 ms
42.1 ms
1850 ms
100x Faster
vs Autoregressive
Direct CWD
(Multiplexer)
Rule Engine
(Regex/Prefix)
Laya MPS
(Apple Silicon)
Laya CPU
(8-Thread)
Autoregressive
LLM (API)
(b) Memory Footprint & Role Decoupling
Memory / VRAM Footprint (MB, log scale)
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
Worker <15MB Zero-Model Invariant
11.2 MB
14.8 MB
1.12 GB
8.30 GB
Worker Node
(Dispatch Core)
Cockpit Node
(Router Core)
Cockpit Node
(+ Laya ModernBERT)
Local 7B LLM
(Autoregressive)
(c) Confidence Calibration & HITL Gate
Decision Confidence Score (%)
100
80
60
40
20
0
95.3%
[Act]
91.9%
[Act]
Action Gate (85.0%)
70.1%
[Quarantined]
HITL Outbox Quarantine Zone (<85%)
32.4%
[Quarantined]
UI Defect
(TerraButton)
Webhook Bug
(Stripe HMAC)
Ambiguous Chore
(Refactor)
Noise Task
(Text anomaly)